Grokking the Advanced System Design Interview

Learn system design through architectural review of real systems.

Level: Advanced | Study Time: 21h | Lessons: 118

33,977 (Students)

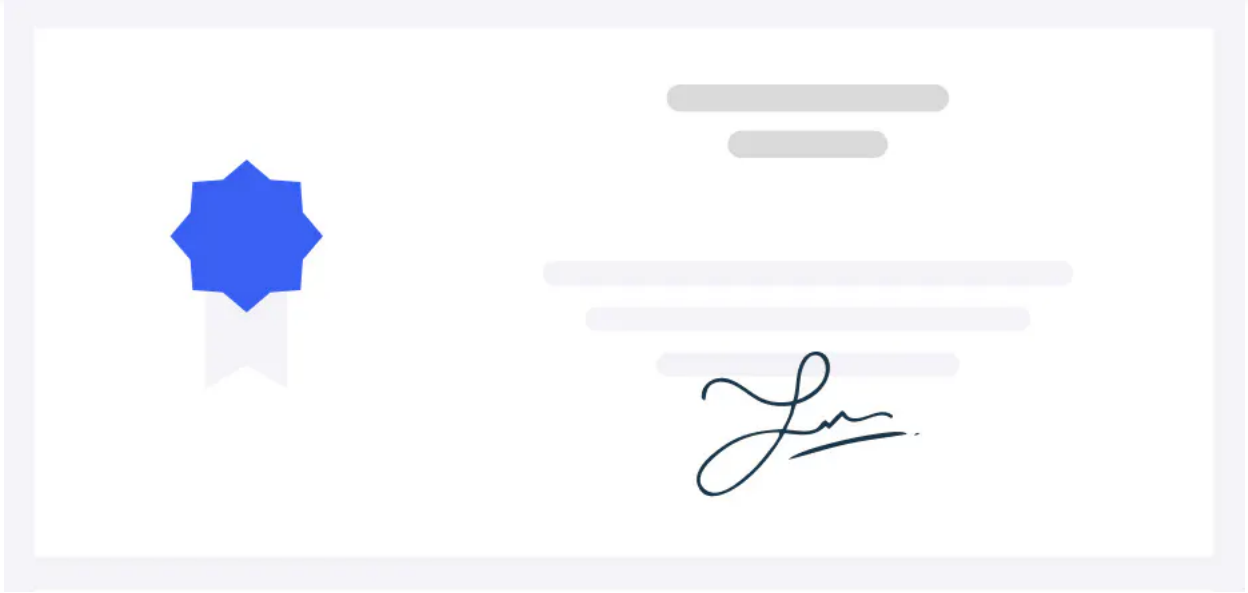
Course Overview

System design questions have increasingly become an integral part of software engineering interviews. For senior engineers, the discussion around system design is considered even more important than solving a coding question. In a system design interview, you can show your real design skills and show how they will work with designing complex systems. It is a given that a good performance in system design interviews will get you a senior position and result in higher salaries.

This course presents the architectural review of famous distributed systems. The main goal is to extract out important design details that are relevant to system design interviews. The course also presents a list of system design patterns that constitute the common design problems and their solutions that different distributed systems have developed over time.



Start Reading



Certification on completing the course.

Download Certificate

Course Content

- What Is This Course About?

2. Dynamo: How to design a key value store? — 12 chapters

- Dynamo: Introduction
- High-Level Architecture
- Data Partitioning
- Replication
- Vector Clocks and Conflicting Data
- The Life of Dynamo's put() & get() Operations
- Anti-entropy Through Merkle Trees
- Gossip Protocol
- Dynamo Characteristics and Criticism
- Summary: Dynamo
- Quiz: Dynamo
- Mock Interview: Dynamo

3. Cassandra: How to Design a Wide-column NoSQL Database? — 12 chapters

- Cassandra: Introduction
- High-level Architecture
- Replication
- Cassandra Consistency Levels
- Gossiper
- Anatomy of Cassandra's Write Operation
- Anatomy of Cassandra's Read Operation
- Compaction
- Tombstones
- Summary: Cassandra
- Quiz: Cassandra
- Mock Interview: Cassandra

4. Kafka: How to Design a Distributed Messaging System? — 13 chapters

- Messaging Systems: Introduction
- Kafka: Introduction
- High-level Architecture
- Kafka: Deep Dive

- Kafka Workflow
- Role of ZooKeeper
- Controller Broker
- Kafka Delivery Semantics
- Kafka Characteristics
- Summary: Kafka
- Quiz: Kafka
- Mock Interview: Kafka

5. Chubby: How to Design a Distributed Locking Service?

14 chapters

- Chubby: Introduction
- High-level Architecture
- Design Rationale
- How Chubby Works
- File, Directories, and Handles
- Locks, Sequencers, and Lock-delays
- Sessions and Events
- Master Election and Chubby Events
- Caching
- Database
- Scaling Chubby
- Summary: Chubby
- Quiz: Chubby
- Mock Interview: Chubby

6. GFS: How to Design a Distributed File System Storage?

15 chapters

- Google File System: Introduction
- High-level Architecture
- Single Master and Large Chunk Size
- Metadata
- Master Operations
- Anatomy of a Read Operation
- Anatomy of a Write Operation
- Anatomy of an Append Operation
- GFS Consistency Model and Snapshotting
- Fault Tolerance, High Availability, and Data Integrity

Criticism on GFS

Summary: GFS

Quiz: GFS

Mock Interview: GFS

7. HDFS: How to Design File Storage System?

12 chapters

Hadoop Distributed File System: Introduction

High-level Architecture

Deep Dive

Anatomy of a Read Operation

Anatomy of a Write Operation

Data Integrity & Caching

Fault Tolerance

HDFS High Availability (HA)

HDFS Characteristics

Summary: HDFS

Quiz: HDFS

Mock Interview: HDFS

8. BigTable: How to Design a Wide Column Storage System?

15 chapters

BigTable: Introduction

BigTable Data Model

System APIs

Partitioning and High-level Architecture

SSTable

GFS and Chubby

Bigtable Components

Working with Tablets

The Life of BigTable's Read & Write Operations

Fault Tolerance and Compaction

BigTable Refinements

BigTable Characteristics

Summary: BigTable

Quiz: BigTable

Mock Interview: BigTable

- Introduction: System Design Patterns
- 1. Bloom Filters
- 2. Consistent Hashing
- 3. Quorum
- 4. Leader and Follower
- 5. Write-ahead Log
- 6. Segmented Log
- 7. High-Water Mark
- 8. Lease
- 9. Heartbeat
- 10. Gossip Protocol
- 11. Phi Accrual Failure Detection
- 12. Split Brain
- 13. Fencing
- 14. Checksum
- 15. Vector Clocks
- 16. CAP Theorem
- 17. PACELC Theorem
- 18. Hinted Handoff
- 19. Read Repair
- 20. Merkle Trees

10. Final Assessment — 2 chapters

- Quiz I
- Quiz II

11. Appendix — 1 chapter

Related Courses





Start Reading

Start Reading

What people say about our courses



Vivien Ruska
Software Engineer

Hey, I wasn't looking for interview materials but in general I wanted to learn about system design, and I bumped into 'Grokking the System Design Interview' on designgurus.io - it also walks you through popular apps like Instagram, Twitter, etc. ✌️

MO JAFRI
Software Engineer

The courses which have "grokking" before them, are exceptionally well put together! These courses magically condense 3 years of CS in short bite-size courses and lectures (I have tried System Design, OODI, and Coding patterns). The Grokking courses are godsent, to be honest.

KAUSHIK JONNADULA
Software Engineer

Thanks for a great resource! You guys are a lifesaver. I struggled a lot in design interviews, and this course gave me an organized process to handle a design problem. Please keep adding more questions.

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